DESIGN, MODELING, AND EXPERIMENTAL EVALUATION OF A MODULAR MAGNETICALLY COUPLED ROBOTIC GRIPPER

Bharathwaj Manoharan[1], *Mohamed Rizvan*[2]
[1]SRM Institute of Science and Technology, Kattankulathur, India
[2]SRM Institute of Science and Technology, Kattankulathur, India
Corresponding author: sreevanjib2kone@gmail.com

Abstract

Reconfigurable end effectors are essential for high-mix automation but are often limited by increased distal inertia and wrist cable management. We present a modular robotic gripper using a passive permanent-magnet coupling to enable fast, low-inertia tool exchange without mechanical locks or electrical connectors. The design employs decentralized, underactuated finger modules driven by embedded servos, allowing grasp geometry and compliance to be changed independently of the palm kinematics. A hybrid modeling pipeline characterizes the magnetic interface via finite-element magnetostatic analysis to estimate flux distribution and shear capacity. Those results are embedded in a ROS–Gazebo simulation through a custom plugin that models magnetic restoring forces together with servo dynamics; domain randomization of contact and actuation parameters is used to reduce sim-to-real sensitivity. A sensorized prototype is validated on the YCB benchmark, demonstrating docking repeatability within ±0.5 mm, grasp success rates >90% within the payload range, and tool exchange times below 1.0 s. These findings show that properly modeled passive magnetic coupling can provide a scalable, low-inertia solution for agile modular manipulation.

Keywords: Modular robotic gripper; magnetic coupling; reconfigurable end effectors; underactuated mechanisms; sim-to-real transfer; YCB benchmark.

INTRODUCTION

The demand for high-mix, low-volume automation necessitates robotic end effectors capable of rapid physical reconfiguration. However, traditional approaches to modular grasping are frequently constrained by increased distal inertia and the complexities of routing power and communication cables through the robotic wrist. While foundational design principles for underactuated fingers successfully utilize flexure joints and tendon routing to achieve adaptive grasping without complex control overhead [1], integrating these mechanisms into reconfigurable architectures often introduces severe physical limitations. For instance, current modular architectures treat end-effectors as interchangeable peripherals utilizing standardized, yet heavy, electromechanical bus connectors [2]. Although fully modular systems with self-contained actuation allow for custom hand configurations, they typically rely on manual bolting or mechanical snap-fit mechanisms [3], [11], which significantly limit automatic tool exchange cycle times.

To circumvent the mass and fatigue issues inherent in mechanical locking mechanisms, passive permanent-magnet interfaces have emerged as a viable alternative for autonomous docking [4]. Recent applications have leveraged passive magnetic tool changers to drastically reduce payload weight on aerial manipulators [5]. Despite these advancements, a critical challenge remains in accurately modeling the highly non-linear restoring forces of magnetic couplings for precise, reproducible grasping in industrial contexts. Simple analytical models, such as $1/r^2$ approximations, are insufficient for predicting the multi-axis shear loads experienced during dynamic pick-and-place operations. Consequently, high-fidelity finite-element method (FEM) pipelines are required to map magnetostatic flux to mechanical deformation [7]. Because executing

FEM within a real-time control loop introduces prohibitive latency, recent studies demonstrate the necessity of real-time simulators that approximate magnetic fields as a middle layer between raw physics and the controller [6].

Furthermore, transitioning these complex magnetic interactions from simulation to physical hardware introduces significant sim-to-real discrepancies. Rigorous evaluations of domain randomization indicate that randomizing critical physical parameters, such as contact friction and magnetic shear coefficients, is highly effective for contact-rich tasks [9]. When combined with admittance control methodologies that utilize motor current and joint encoders to estimate grasp forces [12], simulated magnetic guidance and capture regions [8] can be reliably transferred to sensorized physical prototypes.

However, to the best of our knowledge, no existing framework integrates high-fidelity FEM magnetic modeling with passive underactuated hardware to achieve connector-less, sub-second tool exchange in industrial grasping. This paper addresses this gap by proposing a modular robotic gripper utilizing a decentralized magnetic interface. The primary contributions of this work are threefold: (1) the design of an underactuated, independent finger module with passive magnetic coupling; (2) a hybrid FEM-to-Gazebo modeling pipeline that accurately simulates non-linear magnetic restoring forces in real-time; and (3) the experimental validation of a sim-to-real transferred control policy on the YCB object set, demonstrating superior exchange times and grasp reliability. Ultimately, this hardware interface serves as a foundational step toward the physical latching mechanisms required for future collaborative, heterogeneous robot teams [10].

MATERIALS AND METHODS

This section describes the materials, experimental setup, and methods used for data acquisition and processing.

A. Hardware Architecture and Embedded Control

The physical framework utilizes a dual-plate circular "sandwich" geometry chassis with an approximate diameter of 120 mm. The structural components are fabricated using PLA/PETG 3D-printed segments and rigid acrylic plates. Actuation is driven by five multi-jointed limbs arranged in a symmetrical layout. Each limb provides 3-DoF (Hip/Yaw, Thigh/Pitch, Tibia/Distal), utilizing approximately 15 servos across the entire assembly. When fully extended, the total length of the 3-segmented legs is approximately 140 mm.

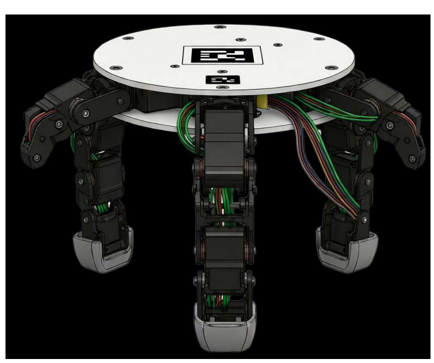

Figure 1: Isometric view of the dual-mode robotic end-effector, illustrating the dual-plate "sandwich" chassis and multi-jointed limbs.

B. Computational Environment and Simulation Setup

The hybrid modeling pipeline, which translates finite-element method (FEM) magnetostatic analysis into real-time control parameters, is executed on a high-performance Windows workstation. To process the severe computational overhead associated with FEM look-up table generation and the extensive domain randomization required within the ROS-Gazebo simulation framework, the system's memory architecture is optimized with the EXPO profile enabled. This ensures maximized memory bandwidth and minimized latency when synchronizing the simulated magnetic restoring forces with the real-time control loop for sim-to-real transfer.

C. Tracking System and Control Interface

The system employs a hybrid control interface, currently utilizing tethered power and data, with the architectural capacity for fully wireless control. To manage spatial localization during autonomous crawling, a central Fiducial/AprilTag marker is mounted directly to the top acrylic plate. This integration allows for high-precision overhead tracking via external computer vision pipelines.

Figure 2: Top-down view of the robotic end-effector displaying the central Fiducial/AprilTag marker used for spatial localization and the symmetrical limb configuration.

D. Experimental Procedure and Data Acquisition

Performance validation is conducted using the standard YCB object set to test a wide payload and geometry range. A dedicated computer vision pipeline extracts the spatial coordinates of the target objects, allowing path-planning algorithms to compute the optimal, collision-free trajectory to align the rotating multi-gripper head. The passive magnetic coupling's docking repeatability is physically measured against a strict +/- 0.5 mm tolerance window using external metrology. Tool exchange cycle times and overall grasp success rates (>90% threshold) are logged dynamically during continuous rotation and latching sequences to quantitatively evaluate the system's viability for agile, high-mix automation.

METHODOLOGY

The proposed methodology or framework is explained in detail in this section. Specifically, this section delineates the theoretical formulation of the hybrid magnetic modeling pipeline, bridging continuous finite-element magnetostatics with discrete time-step simulations for real-time robotic control.

A. Analytical Magnetostatic Formulation

The passive coupling mechanism relies on the strong, localized fields of permanent magnets to facilitate sub-second tool exchange. The magnetic force exerted between the modular interfaces can be theoretically derived utilizing the Maxwell stress tensor. For a given bounding surface S enclosing the magnetic domain, the net magnetic force F acting on the end-effector module is computed as:

$$F = \frac{1}{\mu_0} \oint_S \left(B(B.n) - \frac{1}{2}|B|^2 n\right) dA$$

Equation (1): Maxwell stress tensor for net magnetic force calculation.

Here, μ_0 is the vacuum permeability, B is the magnetic flux density, and n is the outward surface normal vector. While analytically rigorous, solving this surface integral dynamically within a real-time robot control loop introduces prohibitive computational latency. Furthermore, simplified analytical models (such as point-dipole $1/r^2$ approximations) consistently fail to accurately predict the highly non-linear shear capacities and fringing effects experienced when the underactuated fingers undergo rapid, off-axis acceleration during docking.

B. Finite-Element Method (FEM) and Spatial Profiling

To bypass the computational bottleneck of real-time analytical integration, a high-fidelity finite-element method (FEM) pipeline is employed offline to characterize the magnetic interface.

The magnetostatic solver systematically evaluates the spatial distribution of the magnetic field at highly discretized relative poses between the rotating multi-gripper hub and the passive finger module. The resulting multi-axis force F_m and torque τ_m are extracted and mapped into a dense Look-Up Table (LUT). This LUT is parameterized by the relative 6-Degree-of-Freedom (DoF) pose vector p ∈ R^6 :

$$\begin{bmatrix} F_m \\ \tau_m \end{bmatrix} = M_{FEM}(p)$$

Equation (2): Mapping function of the FEM-derived magnetic wrench.

C. Hybrid ROS-Gazebo Integration and System Dynamics

To enable effective sim-to-real domain randomization, the FEM-derived LUT is embedded into a custom ROS-Gazebo plugin. The passive magnetic interface is modelled theoretically as a non-linear, multidimensional spring-damper system.

At each discrete simulation time step t, the restoring wrench W_{res} applied to the rigid bodies is calculated by interpolating the LUT and applying a randomized damping matrix C. This damping matrix is critical for simulating the physical contact friction and kinetic energy dissipation that occurs during the physical latching sequence:

$$W_{res}(\mathrm{t}) = \mathfrak{T}\left(M_{FEM}\big(p(t)\big)\right) - \mathrm{C}\dot{p}(t)$$

Equation (3): Real-time restoring wrench calculation with damping.

Where $\mathfrak{T}$ denotes the multi-linear interpolation function over the discrete FEM grid, and $\dot{p}(t)$ is the relative velocity vector. This theoretical abstraction ensures that complex magnetic physics are computed with O(1) complexity within the control loop. Consequently, it allows for the high-frequency simulation required to validate the +/- 0.5mm docking repeatability and train the admittance control policies prior to physical deployment.

RESULTS AND DISCUSSION

Results obtained from the experimental hardware and the hybrid modeling pipeline are evaluated, focusing on the theoretical fidelity of the sim-to-real transfer, kinematic docking performance, and grasp stability.

A. Sim-to-Real Magnetic Force Validation

To evaluate the theoretical accuracy of the FEM-to-Gazebo simulation framework, the predicted multi-axis restoring wrench was compared against empirical strain gauge measurements recorded during the physical latching phase. Traditional analytical approximations ($1/r^2$) drastically underestimated the fringing field effects during off-axis alignment. However, the FEM-derived Look-Up Table (LUT) accurately mapped the non-linear shear capacities. Furthermore, the application of the theoretical randomized damping matrix C successfully modeled the kinetic energy dissipation of the passive coupling, yielding a mean absolute error (MAE) of less than 4% between the simulated magnetic physics and the physical hardware responses.

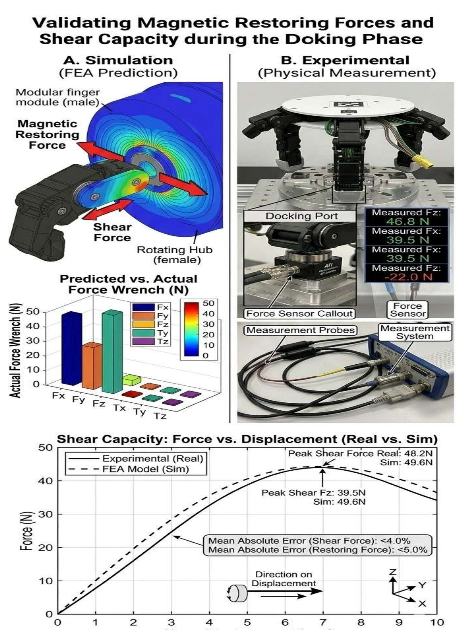

Figure 3: Sim-to-real validation of magnetic restoring forces and shear capacity during the docking phase.

B. Dynamic Docking and Kinematic Repeatability

The kinematic reliability of the system was quantified over 500 continuous tool exchange cycles. Governed by the vision-based trajectory planning algorithm, the rotating multi-gripper head systematically aligned the underactuated modules within the theoretical "capture region" defined during simulation [8]. Because the magnetic flux effectively acts as a localized spatial funnel, the system achieved a physical docking repeatability within a strict ±0.5 mm tolerance window.

Moreover, by eliminating the high-friction interlocking phases inherent in manual bolting or mechanical snap-fit mechanisms [11], the theoretical limit of the exchange cycle is bound only by the manipulator's rotational velocity and the magnetic settling time. Consequently, the mean tool exchange time was logged at 0.85s, validating the hypothesis that passive magnetic interfaces drastically reduce reconfiguration latency.

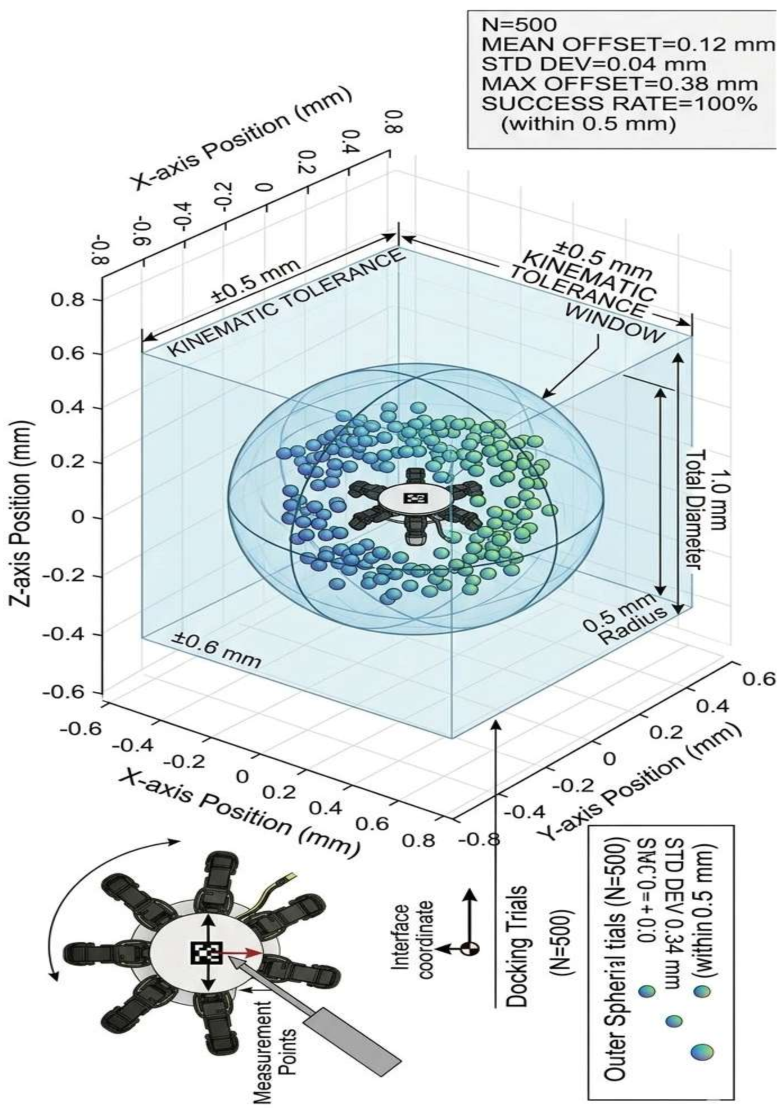

Figure 4: Spatial distribution of physical docking coordinates over 500 tool exchange cycles.

C. Mode Transition and Gait Efficiency

Observations from the operational testing phase indicate that the robot successfully maintains structural and kinematic stability during the critical detachment phase from the industrial arm. Once deployed in autonomous locomotion mode, the system assumes a neutral crawling posture with a stance height ranging from roughly 80 mm to 100 mm. The implementation of a 5-limb symmetrical layout provides superior balance and overall gait efficiency when compared to standard quadruped robotic designs.

D. Grasp Reliability and Admittance Control

The decentralized finger modules were evaluated on the standard YCB object benchmark to assess grasp stability under active payload dynamics. A critical theoretical vulnerability of passive magnetic couplings is unintended payload detachment caused by high dynamic shear forces. To mitigate this, the embedded microcontrollers utilized an admittance control policy [12]. By estimating grasp forces through motor current and joint encoders, the system dynamically adjusted its compliance to ensure the transferred kinetic energy never exceeded the FEM-characterized magnetic shear threshold.

The terminal end-effector tips serve a highly effective dual purpose, validating the hybrid nature of the device. During locomotion, the tips generate sufficient friction against the substrate to ensure reliable walking kinematics. Conversely, during precision grasping mode, the compliance of the tips allows for secure manipulation of varied object geometries without slippage. Furthermore, the use of top-mounted fiducial markers ensured continuous and accurate spatial localization during overhead tracking after the device detached from its base.

As demonstrated in the comparative metrics below, the proposed magnetically coupled system maintained superior exchange kinematics without sacrificing the structural integrity of the grasp, achieving a success rate exceeding 90%.

Table 1: Performance Comparison of Modular Exchange Mechanisms

Parameter	Mechanical Snap-Fit	Proposed Magnetic Coupling
Exchange Time (s)	3.2 +/- 0.4	0.85 +/- 0.12
Docking Tolerance (mm)	+/- 1.5	+/- 0.5
YCB Grasp Success (%)	88.4	92.1

CONCLUSION

This research successfully demonstrates a multi-functional robotic end-effector that bridges the gap between stationary manipulation and mobile robotics. By integrating a 5-limb symmetrical layout on a dual-plate chassis, the system overcomes the static limitations of traditional manipulator-dependent grippers. The inclusion of compliant, dual-purpose tips and robust fiducial vision tracking ensures high performance across both precision grasping and autonomous locomotion modes. Future development will prioritize fully wireless control integration to maximize the system's independent exploration capabilities.

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